NUCLEAR LABORATORY

Fourth Year - First Semester



2026-2027

Name:

Group:

Date:

Name of partner:

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Experiment (1)
Counting Statistics

Purpose

The aim of this experiment is to demonstrate the statistical nature of random events by observing the nuclear radiation for fixed time interval.

Theory

Radiation nuclei appear to decay at random process, each measurement of decay of given nucleus dose not depends on the previous measurement and is not affected by the decay of neighboring atoms. Consequently, any measurement, which is based on observing the radiation emitted in nuclear decay, is subjected to some degree of statistical fluctuation. These inherent fluctuation in nuclear radiation produce on unavoidable source of uncertainty in all nucleus measurements and often can be the predominant source of imprecision or error. Therefore, large number of individual measurements, the deviation of individual count rate from what might be termed the" average count rate" behaves in random manner, small deviation from the average are much more likely than large deviation. The average count rate for (n) independent measurement is given by:

$$\mathrm{N}_{ave} = \frac{\mathrm{N}_1+\mathrm{N}_2+\mathrm{N}_3 \ldots\ldots\ldots\ldots \mathrm{N}_n}{\mathrm{n}} \qquad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

N_1: Count rate for first measurement.

N_2: Count rate of second measurement.

In summation notation N_{ave} can be written as:

$$\mathrm{N}_{ave} = \sum_{i=0}^{n} \frac{\mathrm{N}_i}{n}$$

The deviation of individual count from mean is $(\mathrm{N}_i - \mathrm{N}_{ave})$ from definition of N_{ave} it is clear that

$$0 = \sum_{i=0}^{n} (\mathrm{N}_i - \mathrm{N}_{ave})$$

So the standard deviation (SD)

$$\sigma_{th} = \sqrt{N_{ave}}$$

then,

$$FWHM = 2.35\sigma_{th}$$

And the experimental mean square deviation σ_{exp} evaluated from

$$\sigma_{exp} = \sqrt{\sum_{i=1}^{n} \frac{(N_i - N_{ave})^2}{n-1}} \qquad \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

On the other hand, the statistical theory of error is usually based on the normal (Gaussian) distribution function. The Normal distribution function $W(N_i)$ is continuous function of N_i defined in such a manner that the quantity [$W(N_i)\mathrm{dN}_i$] gives the probability between (N_i and $N_i + dN_i$). The normal distribution function $W(N_i)$ can be written as

$$\mathrm{W}\,(N_i) = \frac{1}{\sigma_{th}\sqrt{2\pi}} \exp\left(\frac{-(N_i - N_{ave})^2}{2{\sigma_{th}}^2}\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots (3)$$

Figure (1) is illustrated the form of the distribution along with its most important properties.

Ideal distribution

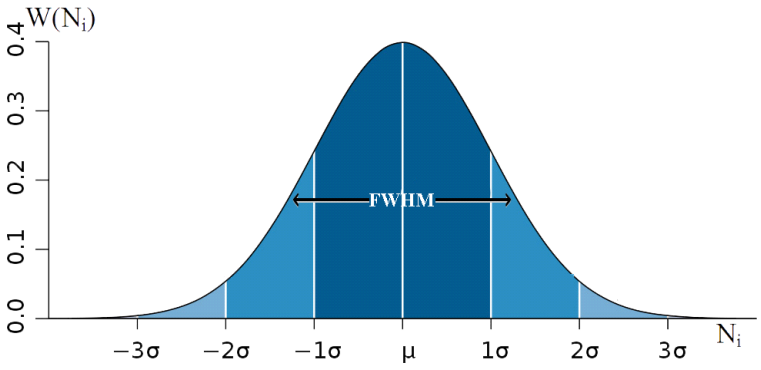


Figure 1: Gaussian Distribution.

Procedure

1. Set up high voltage power supply at the operating voltage and put the radioactive source at few centimeters from center window.

2. Take 100 independent counts for ($10sec$) and record your values in table (1). The scalar value N_i may be directly in the table since for this experiment N_i is defined as

the number of counts recorded for($10sec$) time interval, and then find N_{ave} from eq.(1).

3. Evaluate experiments at (SD) σ_{exp} from eq. (2) and compares it with σ_{th}.

4. Choose a class interval for your reading (not less than ten interval), find the frequency for each interval and draw a histogram it is obtained by drawing rectangles whose height are frequencies and widths are the interval or range

Table (1)

NO.	N_i $(^{count}/_{10sec})$	$N_i - N_{ave}$ $(^{count}/_{10sec})$	$(N_i - N_{ave})^2$ $(^{count}/_{10sec})^2$	$W(N_i)$
1 2 3 4 . . 100				
	N_{ave}	$(N_i - N_{ave})_{ave}$	$\sum(N_i - N_{ave})^2$	

Illustration:

Suppose we take 100 observation of (counting rate per 10 sec) of radioactive source, let this observation being as table (2).The above set of data exhibits no pattern

Table (2)

21	60	70	62	33	85	69	54	82	76
81	31	40	50	25	82	42	51	89	77
77	27	33	43	35	52	63	53	64	76
65	34	44	55	39	58	32	60	56	72
73	45	29	67	46	52	47	65	55	65
77	52	65	76	54	64	79	58	65	99
76	46	57	67	45	65	54	67	58	78
76	59	65	80	75	66	88	56	67	77
87	56	67	87	54	65	57	84	54	65
87	56	64	75	52	65	54	89	64	78

Except the counting rate differ from each other between ($200\ to\ 96$) if we present this data in different way it will provide us with useful information for this we can divide the data into **8** intervals.

Table (3)

NO.	Range of counting rate	Frequency of occurrence of counting rate in that range
1	20……..30	3
2	30……..40	6
3	40……..50	10
4	50…….60	27
5	60…….70	24
6	70…….80	17
7	80…….90	12
8	90…….100	1

The table (3) called frequency table. Plot a graph between the range of counting rate and Frequency of occurrence of counting rate in that range, as shown in figure (2).

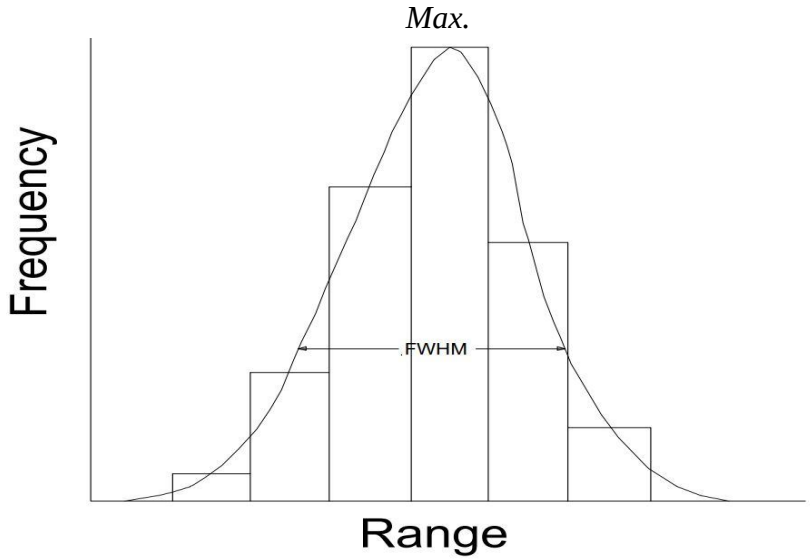


Figure (2)

Questions:

1- Define standard deviation (S.D) and $FWHM$.

2- What is the Gaussian distribution? And why we are use it?

3- Why we use the statistical method to record the nuclear radiation (counts)?

4- There are another function (method) for determining the distribution function, write down them.

5- What is the useful of this experiment?

6- What is the main error of this experiment?

7- Write down the physical main distributions with their equations.

Experiment (2)

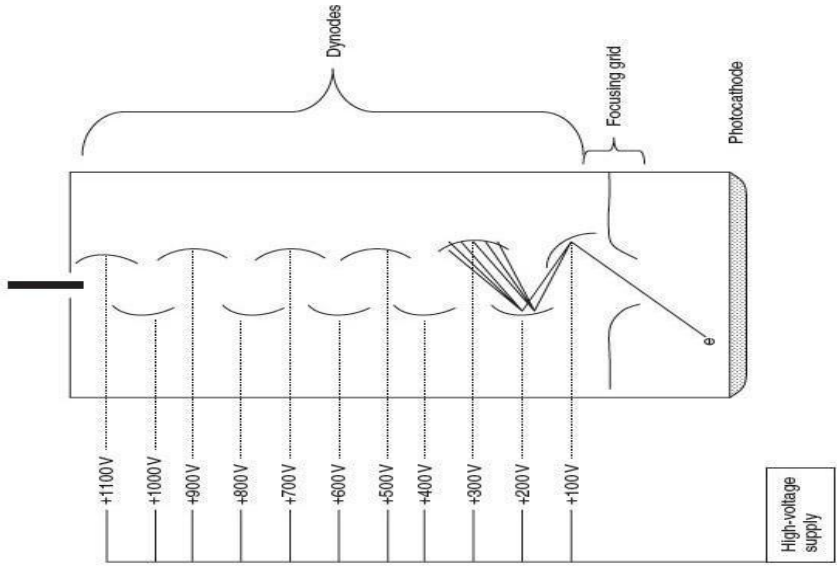
Determination of the operating voltage of the scintillation detector

Purpose:

This experiment aims to determine the operating voltage of the scintillation detector.

Theory:

The thallium-activated sodium iodide detector or $NaI\ (TI)$ detector responds to γ-ray by producing a small flash of light, a scintillation. After the gamma-ray penetrates the active medium and interacts with the crystal, the electrons of the crystal are either ionized or excited; the electrons jump from the valence band to the conduction band. As we know, each level has a mean life, and after this time, the electron returns to the valence band and emitting a photon in the energy range of ultraviolet or visible light. When the photons are emitted, they hit the photo-cathode layer (the first part of the photomultiplier tube). Here, the photons are again changing for the electrons by the photoelectric effect. The second part of the photomultiplier tube is the dynodes, consisting of a composite metal plate and a large number of free electrons on the metal. When the high voltage is applied on these dynodes, the electrons are pulling by the dynodes, then the electrons hit the dynodes, and each electron removes several electrons in each dynode. As explained in the figure below each dynode applied a specific voltage, the first dynode applied low voltage, the second applied more voltage than the first, and so on until the last dynode, as shown in figure(1). In this experiment, we are changing the high voltage, which means changing the dynodes voltage. Changing the dynode's voltage causes to change in the number of electrons which reaches the pre-amplifier.



Figure(1)

Procedure:

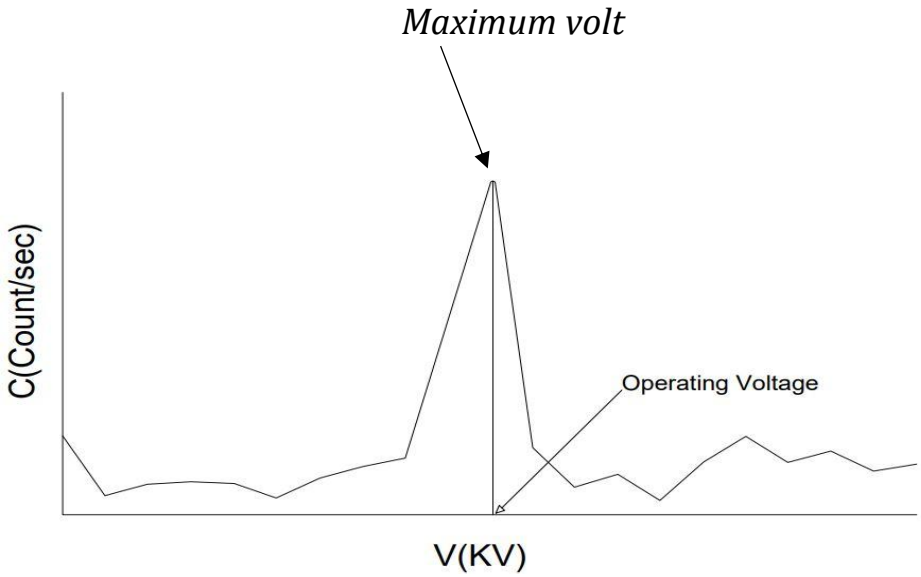
1- Turn on the Scintillation detector. Place the Na^{22} source or another source far enough away from the window of the detector.

2- Taking the data by changes the high voltage from the ($0.80\ Kev$) to ($1\ Kev$) and read the count rate for $10\ second, N$ (count/10sec).

3- Remove the source and take the background counting rate N_B $(Count/10\ Sec)$ at the same values of high voltage as in step (2) above.

V (kV)	N $(Count/10\ Sec)$	N_B $(Count/10\ Sec)$	$C = N - N_B$ $(Count/\text{sec})$
0.80			
0.81			
.			
.			
1			

4-Plot a graph between voltage and correction (C) as shown in fig (2).



Figure(2).

Questions:

Physically and mathematically discuss the following questions:

1. What is the operating voltage?
2. What is the difference between the operating voltage and baseline voltage?
3. What is the effect of changing voltage on the dynodes?
4. Is the changing of operating voltage affected on the widen or narrow of the gamma spectrum peaks?
5. How the scintillation detector converts the $\gamma - ray$ energy to pulse voltage?
6. Interpret the diagram (graph) between the count rate and the high voltage.
7. Explain the role (work) of dynodes in the $NaI(Tl)$ detector.
8. What is the usefulness of this experiment?
9. What is the primary error of this experiment?

Experiment (3)

Energy calibration of the scintillation on spectrometer and energy determination of the unknown source

Aim

The aim of this experiment is to determine the energy of unknown source by using known source and to determine the resolution of your detector which is used for this experiment.

Theory

The decay process of radioactive nuclei results in the emission of nuclear radiation, such as Alpha, Beta, Neutron particles, and Gamma-ray. Often the decay is to an excited state of daughter nucleus, which usually decay by emission of gamma ray, the energy levels (decay scheme) and therefore the gamma-ray energies for any nucleus is unique and can be used to identify the nucleus. To use γ-ray spectrometer to find the energy of any unknown γ-rays, the spectrometer should be calibrated with standard radio-active source such as ($Co^{60}, Cs^{137}, Na^{22}$) which their energies were known previously. The energy levels and decay schemes of these sources are given by below:- We found that Co^{60}, decays by electron emission to the second excited state of Ni^{60} which de excited by emitting two successive γ-rays of energies ($1173KeV\ and\ 1333KeV$), while Na^{22} decays by position emission to first excited state of Na^{22} which is de-excited by emitting gamma-ray with energy $1275KeV$.

A modern digital gamma-ray spectrum is in essence a list of numbers of pulses measured with in small consecutive Pulse height ranges. Detector calibration allows the gamma-ray spectrum to be interpreted in terms of energy, rather than channel number or voltage, and amount of radionuclide, rather than number of Pulses. In addition, especially if computer analysis of the spectrum IS completed, it may be

necessary to Provide information about the Peak width variation with energy or channel number, there are three main calibration tasks:

- Energy calibration _the relationship between channel and energy.
- Peak width calibration _the variation of Peak width with energy.
- Efficiency calibration _ the relationship between number of counts and disintegration rate.

Part A:
Energy calibration

The object of energy calibration is to derive a relationship between Peak Position in the spectrum and the corresponding gamma-ray energy. This is normally Performed before measurement ,if only in a preliminary manner, but it is usual for spectrum analysis programs to include more sophisticated calibration options .Energy calibration is accomplished by measuring the spectrum of a source emitting gamma-rays of precisely known energy and comparing the measured Peak position with energy.

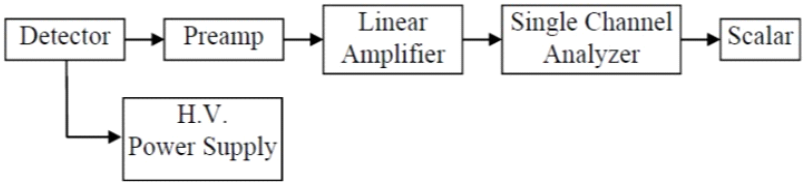
Part B
Resolution

The quality of a given scintillation counting system is often characterized by the magnitude of its energy resolution. The system resolution is a measure of the Photo Peak sharpness.

Resolution is defined as the ratio of the full width at half maximum (FWHM) of the full energy Peak (called the Photo Peak when dealing with Photo detectors) to the energy midpoint of the full energy Peak. Thus, the resolution is dimensionless fraction conventionally expressed as a percentage. With a small resolution, the detector can separate two radiations whose energies are close together.

Procedure

1. Set up the electronic equipment as shown in the figure (1).



Figure(1)

2. Place the Na^{22} radioactive source at about (3) cm far from the detector.
3. Set operating voltage at$(0.9\ kV)$.
4. The first step of this experiment involves taking the data between ($BLV\ volt$) and($N\ count/10sec$). Tabulate the data as shown below.

Table 1

BLV (Volt)	N (count/10sec)
0.02 0.04 0.06 . . . . Certain range	

And for the second Part of this experiment Place Co^{60}radioactive source at about (3 cm) far from the face of detector. Take the data between BLV and count Per 10 second. Table of data as shown below.

Table 2

BLV (Volt)	N (count/10sec)
0.02 0.04 0.06 . . . Certain range	

5. Discuss the result physically after completing the graph and calculation steps.

Graph

First graph between N $(count/10sec)$ and BLV, as shown in Fig. (2).

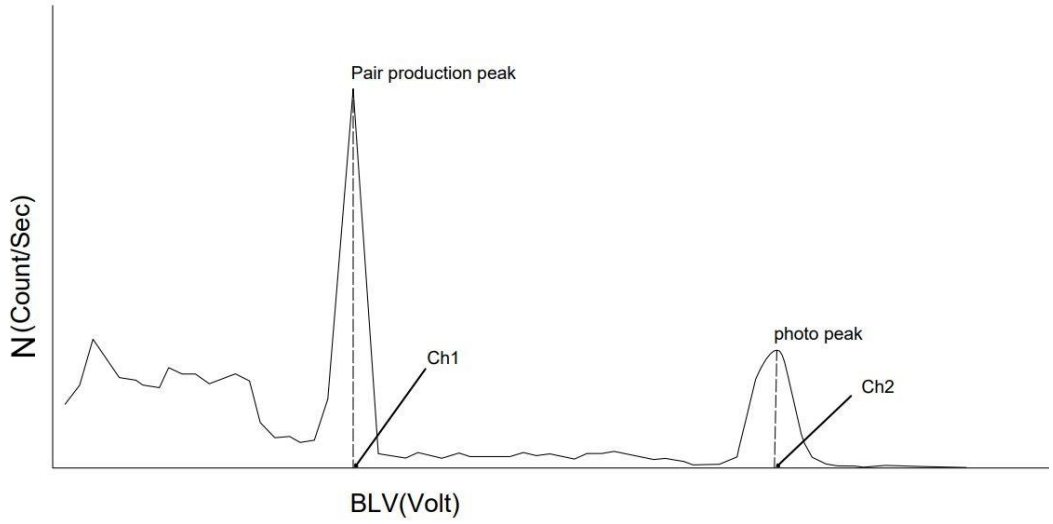
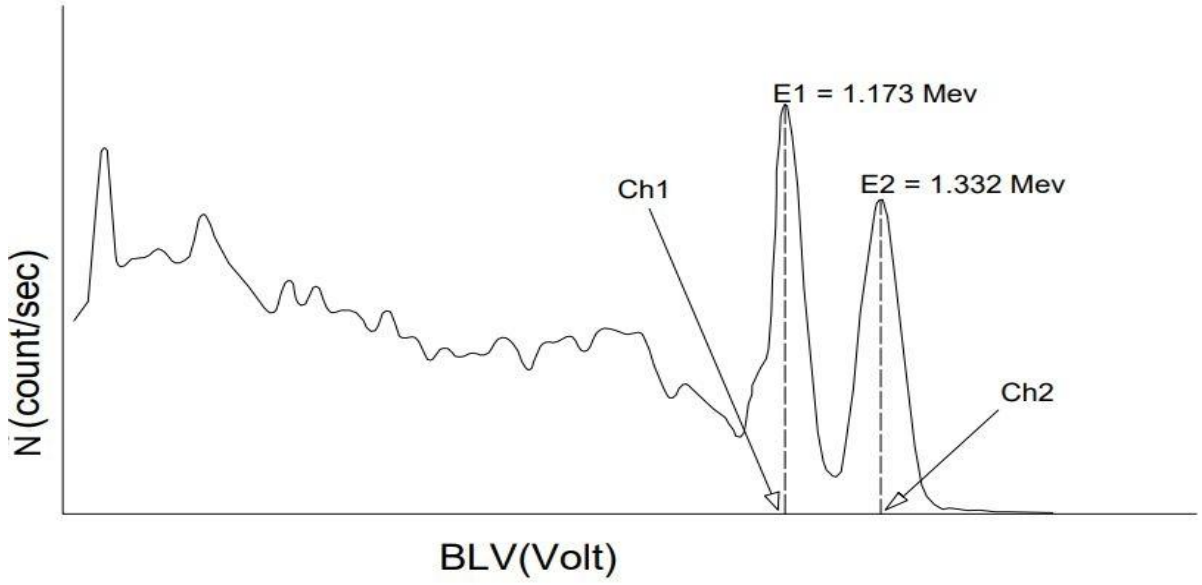


Figure 2

The value of ch_1 and ch_2 must be determined on the graph and its used into calculation. Second graph, from a table-2 Plot a graph between N (count/10sec) and BLV, As shown in the Fig. (3).



Figure(3)

NOTE: The value of ch_1 and ch_2 must be determined on the graph, and its used into calculation**.**

Calculation

Part A

Note figure 3 is the spectrum of Co^{60} radioactive source in this experiment the energy of Peak are known ,then by this Procedure its used to determine the unknown energy Peak of Na^{22} spectrum.

From the fig. 3 we can determine the value of ch_1 ,ch_2,E_1 and E_2 and it's used into law of calibration factor,

Calibration factor for standard source

$$Co^{60}(\text{c.f}) = \frac{E_1 - E_2}{ch_2 - ch_1} \text{in } (\frac{M\ ev}{V}) \text{ .....(1)}$$

$$E_{Peak}\ (\text{Mev}) = ch_{Peak}\ (\text{v}) \times (\text{c.f})\ (\frac{M\ ev}{V}) \text{ .....(2)}$$

And from Fig .2 we can determine the value of ch_1 and ch_2, then by using equation (2) we can calculate the energy of each Peaks ($E_{photo\ Peak}$ and $E_{Pair\ Production}$) .

$$E_{photo\ Peak} = ch_2 \times c.f$$
$$E_{Pair\ Production} = ch_1 \times \text{c.f}$$

Draw the graph between the energy of the Photo-Peaks and their channel numbers.

Part B

The resolution of Photo Peak for Na^{22} is

$$\text{R} = \frac{FWHM}{ch_2} \times 100\%$$

Where ch_2 is the channel number for the maximum count for the Photo Peak on the figure 2 and the FWHM determined from the same graph for the Photo Peak.

And the resolution of photo Peak for Co^{60} is

$$\text{R} = \frac{FWHM}{ch_2} \times 100\%$$

Where $ch_2$is the channel number for the maximum count for the Photo Peak on the figure(3) and the FWHM determined from the same graph for the Photo Peak.

Questions

1. What is the calibration factor, and why the calibration is important?
2. Is calibration factor constant for any instruments?
3. Why the peaks broaden of a linear line?
4. What is the resolution? Why the resolution is important?
5. How the resolutions change with the FWHM?

Experiment (4)

Basic Identification in Electronic Measurement Systems

Purpose

The purpose of this experiment is to train the students with the basic techniques to use to checking for proper system arrangement and responses of the nuclear devices (apparatus).

Apparatus

$NaI(TI)$ crystal and phototube assembly, Photomultiplier tube base, Preamplifier, Main amplifier, Single channel analyzer (SCA), Co^{60} (initial radioactivity of the source was 74 KBq on 4 September 2004), High voltage power supply voltage, NIM Bin and power supply, Stand for NaI(TI) detector, Cable set and Oscilloscope.

Description of the electronic circuits

1. The oscilloscope

The used oscilloscope in this experiment helps us to observe the input and output pulse for the various modules in the system and to determine whether the wave-shapes, amplitudes, and timing are correct with respect to the rest of the equipment. Most oscilloscopes in nuclear laboratories have about 30 knobs and adjustments for performing the various functions for which they were designed. Fortunately, only about 10 of these parameters are necessary for observing input and output pulses from modules or for making simple timing adjustments.

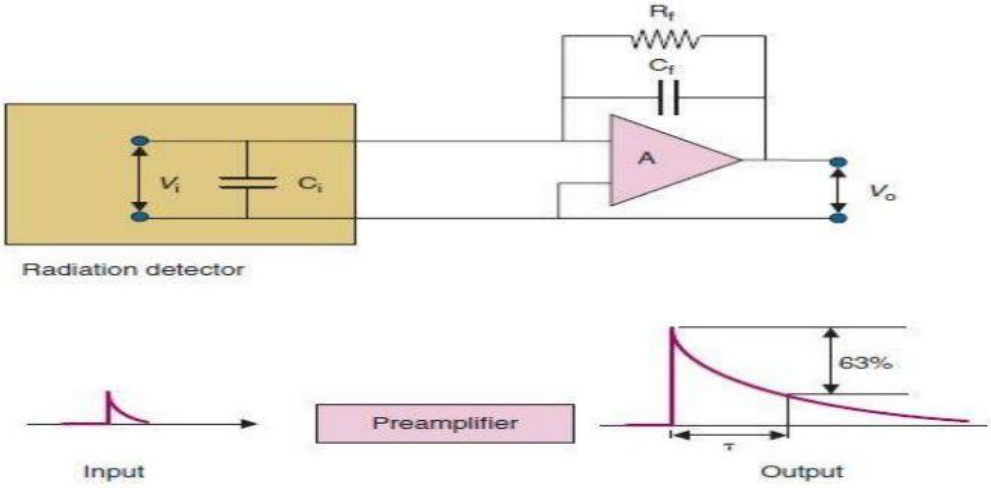
2. $NaI(Tl)$ Scintillation detector

Several different radiation detectors are used in the various projects in the nuclear physics laboratory. In this experiment we use a Sodium Iodide scintillator with

Thallium impurities acting as the activator, $NaI(Tl)$. It is used for the detection of x-ray and gamma-ray radiation with medium resolution in such a way that the photoelectric effect or Compton scattering in the crystal lead to scintillation light that is usually converted to an electric pulse by photomultiplier tube.

3. Preamplifier

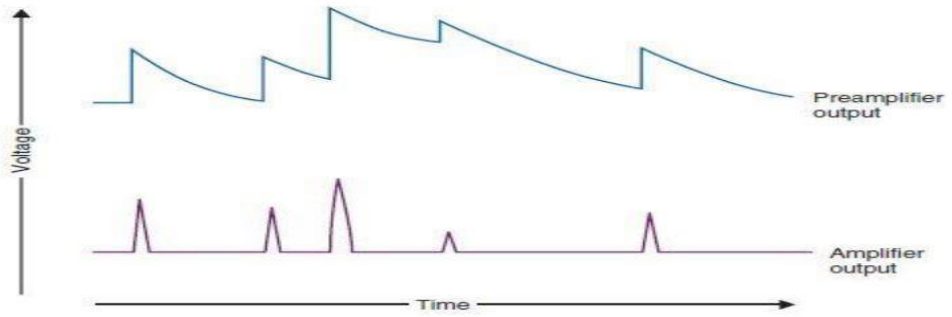
The output generated by the detector is a small charge pulse. It is necessary to precondition this pulse so that it can be utilized in a linear amplifier system. The linear amplifier system provides the necessary pulse shaping and amplification required for use with the measuring unit. A preamplifier is generally inserted immediately after the detector to provide integration of the charge output of the detector into a usable voltage pulse. The output of the preamplifier is connected to pulse-shaping main amplifiers. Impedance matching between the detector and the pulse-shaping and amplifier stages that follows is provided by the preamplifier. Hence a preamplifier is a device which provides impedance matching, amplification, and often, charge integration of a detector signal before transmission to the main amplifier.



4. Main amplifier

The main amplifier further increases the signal from the preamplifier so that it can be used within the dynamic operating range of the system. Linear amplifiers are required to ensure that the relationship between the input and output signals is

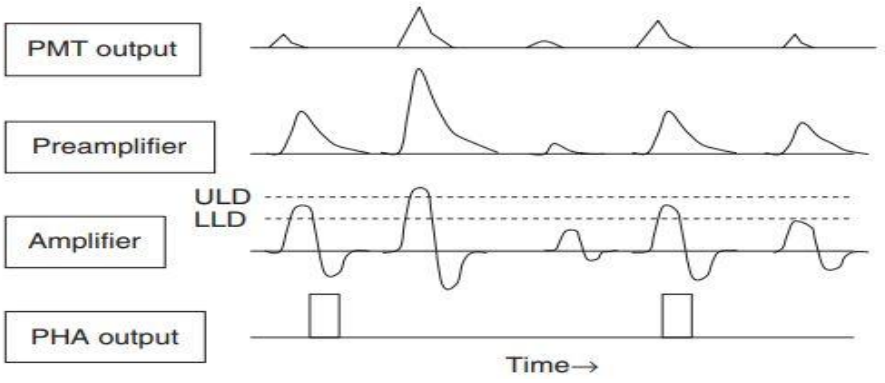
maintained. Hence, an amplifier is a device, whose output signal is greater than, but proportional to, its input signal. The gain of the amplifier is then the ratio of the output signal amplitude to the input signal amplitude. The amplifiers provide two types of output signals, known as unipolar (either positive or negative) and bipolar (having both positive and negative positions).The unipolar pulse is preferred for high resolution, while the bipolar pulse is useful for timing applications and high count rate situations.



5. Discriminators and pulse height analyzers (Single channel analyzer, SCA)

An integral discriminator accepts, as logic pulse input, the varying height pulses from a main amplifier. It yields a standardized a logic pulse output for each input pulse whose amplitude exceeds the discriminator level setting. This level ranges from a minimum level (often 0.1 V) to a maximum level (generally 10 V). This transformation from linear to logic pulse serves two purposes. First, the integral discriminator presents easily counted logic pulses to the following pulse counter. Second, the integral discriminator excludes from analysis all pulses below a region of interest. Nose pulses (i.e., unwanted disturbances transmitted with or added to the wanted signal by the processing circuit) in the main, small in amplitude and should be excluded. The single channel pulse height analyzer (SCA) permits more complex analysis of incoming pulses before pulse counting. Each pulse is compared to two voltage (energy) levels; only those which are greater than the lower level and are less than the upper level generates the logic pulse to be counted. The upper level control may operate in two

ways: either as an independently set upper limit or as a window of a given width opened on top of the lowest level. A single channel analyzer is normally used to establish a window; it yields a logic pulse output only when an incoming linear signal is within its window:



6. Scalars

The output logic pulses from the integral discriminator or (SCA) are routed to a pulse counting device, called a scalar. Modern scalar count accumulated pulses and displays the running total in decimal format.

7. Timers

The duration of a pulse-counting experiment is normally determined by a timer. In the simplest case, this might be a stopwatch, but good experimental procedure dictates the use of an electronic timer for determining experiment duration. In this manner, successive counting times can be held constant.

Procedure

1. Connect the attenuated output of the preamplifier to the scope. Observe that the output pulse at a fixed input pulse. Draw the pulse shape (the signal) on a graph paper with identical scales of the oscilloscope.

2. Connect the output of the preamplifier to the input of the amplifier. Use unipolar output to the oscilloscope. Observe the amplitude of the output pulse shape at a fixed

input pulse and a fixed gain on the scope. Change the toggle switch to bipolar and observe the pulse shape at a fixed input pulse and a fixed gain. Draw the pulse shape on a graph paper with identical scales of the oscilloscope.

3. Connect the output of the amplifier to the input of the SCA and observe the output pulse shape of the SCA on the scope.

4. Determine time period (τ) for each above step, and then calculate the frequencies of them.

Questions

1. Explain the meaning of pulse shape for the following device
(Preamplifier, Amplifier and Single channel analyzer).

2. Compare between single channel analyzer and multichannel analyzer electrically and physically.

3. Compare the amplitude and time of pulse of the Preamplifier, with main amplifier and Single channel analyzer.

4. If possible, investigate the function (work) of each deice via the graphs?

5. What is the useful of this experiment?

6. What is the main error of this experiment?

EXPERIMENT (5)

DETERMINATION OF THE DEAD TIME (RESOLVING TIME) OF A G.M. COUNTER BY TWO SOURCE METHOD

Purpose:

The aim of this experiment is to find the dead time of the Geiger-Muller detector and actual counts.

Theory:

The time following the ionizing event's entry in the counter during which the detector remains insensitive to the next event is called the dead time (T) of the counter. This arises from the slow motion of the positive ion sheath from the anode wire to the cathode. The presence of the positive ion cloud in the vicinity of the anode lowers the electric field to such value that the required size pulse will not be formed if another particle entered the counter soon after the first one and is therefore likely to be missed. After some time, the positive ions reach the cathode, and the counter covered entirely to receive another particle and develop a pulse of standard size.

The time of travel of positive ions from the anode to cathode is, in principle, the dead time of the counter, but if the input sensitivity of scalar is higher so that it can also register pulses of less than normal, the dead time correspondingly smaller because in this case, the next event does not need to wait for the positive ions to go to the cathode.

Therefore, it is clear that the dead time of the counter depends on the kind of scalar used in the experimental set-up and the voltage applied to the anode. The resolving time can also be defined as the minimum time required by the set-up to resolve two successive ionizing events entering the counter.

Procedure:

1- Place the first source (Sr^{90}) at an incident angle (45°) with the projection of the detector, and locate the source ($4\ or\ 5\ cm$) away from the detector. Record the count rate (n_1) ten times for ($10\ sec$).

2- Place the second source (Sr^{90}) on the other side with the same geometry and determine the count rate (n_2) as in step 1.

3- Place both sources and record (n_{12}) as in step 1.

4- Find the background count rate (n_b) for ($10\ sec$).

Steps 1, 2, 3, and 4 present in the table below:

$n_1(Count/10\ sec\ )$	$n_2(Count/10\ sec\ )$	$n_{12}(Count/10sec)$	$n_b(Count/10sec)$
$Ave.of(n_1)$	$Ave.(n_2)$	$Ave.(n_{12})$	$Ave.(n_b)$

Calculations

We can evaluate the dead time of G.M. counter by two source method if we assume that:

n_1: is the count rate of the first source with background

n_2: is the count rate of the second source with background

n_{12}: is the count rate of the two sources with background

n_b: is the count rate of the background only

$$T = \frac{N_1+N_2-N_{12}}{2N_1N_2} \quad \text{.....(1)}$$

The resolving time (T) is then given by

Also, we can calculate the dead time by this equation:

$$T_1 = \frac{N_1N_2-[N_1N_2(N_{12}-N_1)(N_{12}-N_2)]^{1/2}}{N_1N_2N_{12}} \quad \text{..... .....(2)}$$

Where

$N_1 = n_1 - n_b$, $N_2 = n_2 - n_b$ and $N_{12} = n_{12} - n_b$

N_1 for one second $= \frac{N_1}{10}$, $N_2$for one second $= \frac{N_2}{10}$, and N_{12} for one second $= \frac{N_{12}}{10}$

The value of N_1, N_2, N_{12} is used to determine the dead time.

Then the actual count or the source counting rate (n) is given by:

$$n = \frac{N}{1-NT} \quad \text{.....(3)}$$

$n_1 = \frac{N_1}{1-N_1T}$, $n_2 = \frac{N_2}{1-N_2T}$ and $n_{12} = \frac{N_{12}}{1-N_{12}T}$

Compare result of equation (1) with other of equation (2).

Questions:

Physically and mathematically discuss the following question:

1- Define the Dead time (resolving time) of the detector?

2- What is the actual count?

3- Is there any difference between the dead time for the Scintillation detector and the Geiger Muller counter? Or why determining Geiger Muller detector dead time is important?

4- Explain which of these equations are better for evaluating the dead time?

5- What is the usefulness of this experiment?

6- What is the primary error of this experiment?

7- Why the value of dead time is very small?

Experiment (6)

Efficiency of Geiger Muller Counter

Apparatus

Geiger-Muller tube, radioactive source Sr (β -source) in supporting base, forceps for handling source, scalar, clamp and stand, stop-watch, meter rule.

Theory

All radiation detectors will, in principle, give rise to an output pulse for each quantum of **radiation** that interacts with in its active **v**olume. For Primary charged radiation such as alpha or beta particles, interaction in the form of ionization or excitation will take place immediately upon entry of the particle in to the active volume. after traveling a small function of its range, a typical particle will form enough ion pairs along its path to ensure that the resulting pulse is large enough to be recorder. Thus, it is often easy to arrange a situation in which a detector will see every alpha or beta particle that enters its active volume. Under these conditions, the detector is said to have a counting efficiency of 100%. On the other hand, uncharged radiation such us gamma rays or neutrons must first undergo a significant interaction in the detector before detecting is possible. Because these radiations can travel large distances between interactions, detectors are often less than 100% efficient .It then becomes necessary to have a precise figure for the detector efficiency in order to relate the number of pulses counted to the number of neutrons or photons incident on the detector. It is convenient to subdivide counting efficiencies in to two classes: ***absolute and intrinsic***

Absolute efficiencies are defined as

$$\epsilon_{abs}=\frac{number\ of\ count\ recorded}{number\ of\ radiation\ quanta\ emitted\ by\ sourse}$$

They are dependent not only on detector properties but also on the details of the counting geometry (primarily the distance from the source to the detector). The intrinsic efficiency is defined as

$$\epsilon_{int} = \frac{number\ of\ count\ recorded}{number\ of\ radiation\ quanta\ incident\ on\ detector}$$

And no longer includes the solid angle subtended by the detector as an implicit factor. The two efficiencies are simply related for isotropic sources by

$$\epsilon_{int} = \epsilon_{abs} \times \frac{4\pi}{\Omega}$$

Where Ω is the solid angle of the detector seen from the actual source position.

$$\Omega = \frac{\pi r^2}{d^2}$$

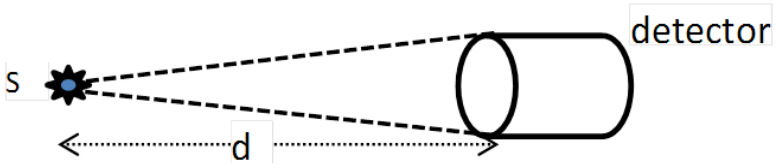


Figure 1: G.M Detector and Source Geometry Diagram

It is much more convenient to tabulate values of intrinsic rather than absolute efficiencies because the geometric dependence is much milder for the former .The intrinsic efficiency of a detector usually depends primarily on the detector material, the radiation energy, and the Physical thickness of the detector in the direction of the incident radiation. A slight dependence on distance between the source and the detector dose remain however, because the average path length of the radiation through the detector will change somewhat with this spacing.

The intrinsic efficiency is

$$\epsilon_{int} = \frac{4d^2 N'}{N(t) r^2}$$

Where$(r = 0.675cm) is$ the radius of the tube window.

And where $N'$is the actual count

$$N' = \frac{N - N_b}{1 - (N - N_b)T}$$

The dead time of detector($T = 400\mu s$).And the$N(t)$ is the total number of particles emitted by the radioactive source per unite time.
At the time $(t)$of experiment from the date of production of radioactive source, we have

$$N(t) = N_0 e^{-\lambda t}$$

Where $\mathrm{N}(0) = N_o$ is used to the initial radioactive nucleus at time $(t = 0)$ and(λ) is the decay constant that is related to the half-life as below

$$\lambda = \frac{0.693}{T_{\frac{1}{2}}}$$

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Procedure

1. Place the β source (Sr^{90}) at (3 cm) from the tube windows.

2. Record the count rate $N\ per\ 10\ sec$.

3. Record the background count rate $N_b\ per\ 10\ sec$.

4. Repeat each measurement in (2) and (3) 10 times to find the average.

5. Use $N_o\ = 74000$, the time of manufacture source (t) is the (3/9/2004), half life time ($T_{1/2}$) for (Sr^{90}) is equal to (28.5 year) and then find$\boldsymbol{\epsilon_{int}}$.

6. Same steps repeated for $\gamma -$ ray energy source (Na^{22}). Use N_0=74000 ,the time of manufacture source (t) is the (3/9/2004), half life time ($T_{1/2}$) for (Na^{22}) is equal to (2.6 $year$).

Questions

1. Define efficiency of detector.
2. What are the Parameter which the efficiency of the detector depend on?
3. Classify the types of efficiency and on what they depending?
4. Is the efficiency related with the detector or the source?
5. For what type of radiation Geiger Muller is more sensitive why?
6. Determine range of efficiency value? Can we increase the efficiency?
7. Define and explain geometric factor meaning.
8. Convert last equation of efficiency to formula which contain geometrical factor.
9. What is the useful of this experiment?
10. What is the main error of this experiment?